

PROJECT ON NETWORK TRAFFIC ANALYSIS AND THREAT DETECTION USING WIRESHARK

1. Title of the Project

Network Traffic Analysis and Threat Detection Using Wireshark

2. Objective

The goal of this project is to analyze and monitor network traffic between virtual machines in a controlled lab environment. By capturing and inspecting packets exchanged between a Kali Linux attacker machine and a Windows 11 target machine, the project aims to understand how network communication works, identify potential vulnerabilities, and develop skills in traffic analysis using industry-standard tools.

3. Tools and Technologies Used

- Kali Linux (attacker machine)
- Windows 11 (target machine)
- VirtualBox
- Wireshark (packet capture and analysis)

4. Environment Setup

The lab uses a Host-Only network adapter to isolate traffic between two virtual machines:

- Kali Linux — IP Address: 192.168.56.101
- Windows 11 — IP Address: 192.168.56.1

All traffic is contained within the isolated virtual network, ensuring a safe and repeatable lab environment with no external exposure.

5. Methodology

Step-by-Step Process

Step 1 – Setting Up the Lab Environment

- Installed and configured Kali Linux and Windows 11 virtual machines using Oracle VirtualBox.

- Connected both virtual machines through a Host-Only Network Adapter to create an isolated lab environment.

Step 2 – Capturing Network Traffic

- Opened Wireshark on the Kali Linux virtual machine.
- Selected the Host-Only network interface to capture only the traffic exchanged between the two virtual machines.
- Started packet capture before generating any network activity to ensure all packets were recorded.

Step 3 – Generating Network Activity

- Performed a ping test from Kali Linux to Windows 11 to generate ICMP traffic.
- Allowed normal network communication to take place so that different types of packets could be captured during the session.

Step 4 – Host Discovery Using Ettercap

- Opened Ettercap on the Kali Linux virtual machine.
- Scanned the Host-Only network to discover active devices.
- Verified that both Kali Linux and Windows 11 were successfully detected in the Ettercap Host List before proceeding with further analysis.

Step 5 – Demonstrating ARP Poisoning

- Opened Ettercap
 - Sniff → Unified Sniffing
 - Select → eth0
 - Hosts → Scan for Hosts
 - Hosts → Host List

Added,

- Windows machine as Target 1
- Gateway/Kali as Target 2

Started Mitm → ARP Poisoning

Enabled Sniff Remote Connections

Wireshark captured the ARP packets.

6. Features Implemented

- IP address configuration for isolated VM-to-VM communication
- Live packet capture using Wireshark on the host-only network interface
- Identification of ARP request patterns

7. Results

- Successfully established bidirectional communication between Kali Linux (192.168.56.101) and Windows 11 (192.168.56.1).
- Detected open TCP ports on the Windows 11 machine, indicating active services.
- Observed ARP traffic confirming MAC address resolution on the local subnet.

8. Challenges Faced

- Network Interface Identification — Determining the correct interface in Wireshark for the host-only adapter required checking VirtualBox network settings.
- Windows Firewall Blocking Traffic — ICMP packets were initially blocked by the Windows 11 firewall; resolved by creating an inbound rule to allow ICMP.
- IP Addressing Conflicts — Ensured both VMs were on the same subnet (192.168.56.0) to avoid routing issues.

9. Conclusion

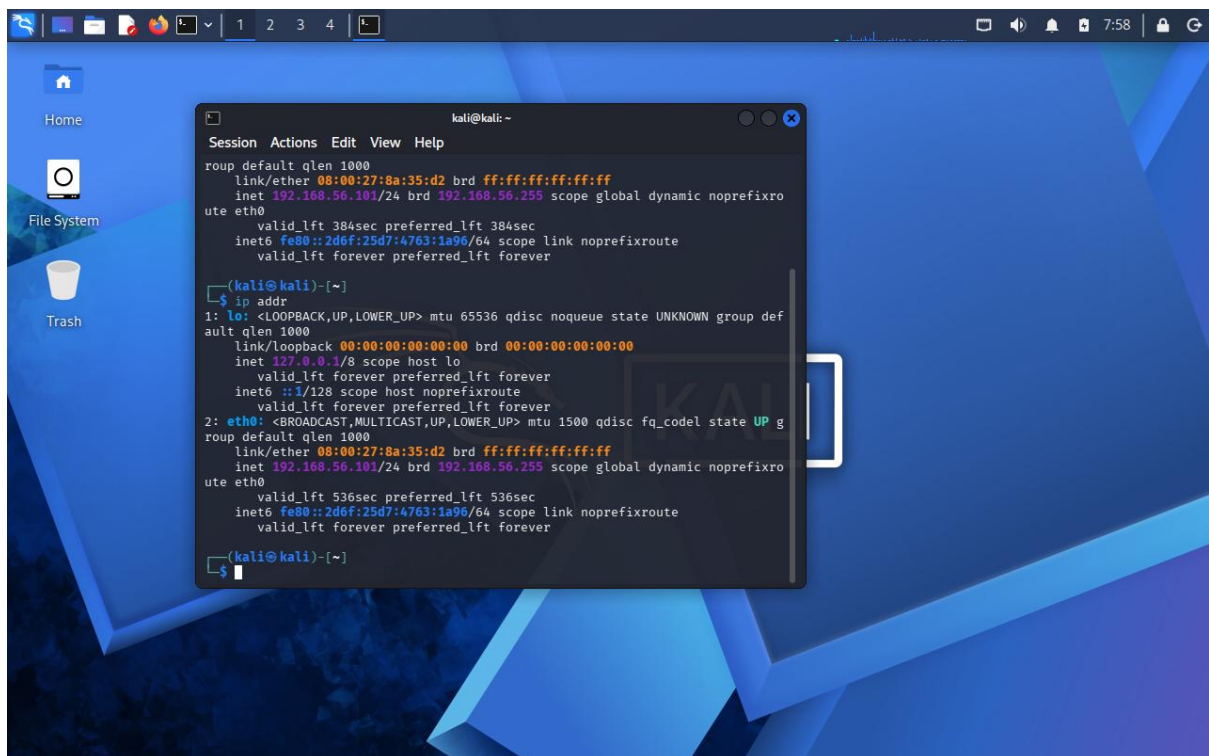
This project provided hands-on experience in setting up an isolated virtual network environment and performing real-time traffic analysis. It deepened understanding of fundamental networking protocols (TCP/IP, ICMP, ARP), packet-level communication, and the role of tools like Wireshark and Nmap in network security analysis. The skills developed are directly applicable to real-world network monitoring, intrusion detection, and cybersecurity incident response.

10. Screenshots

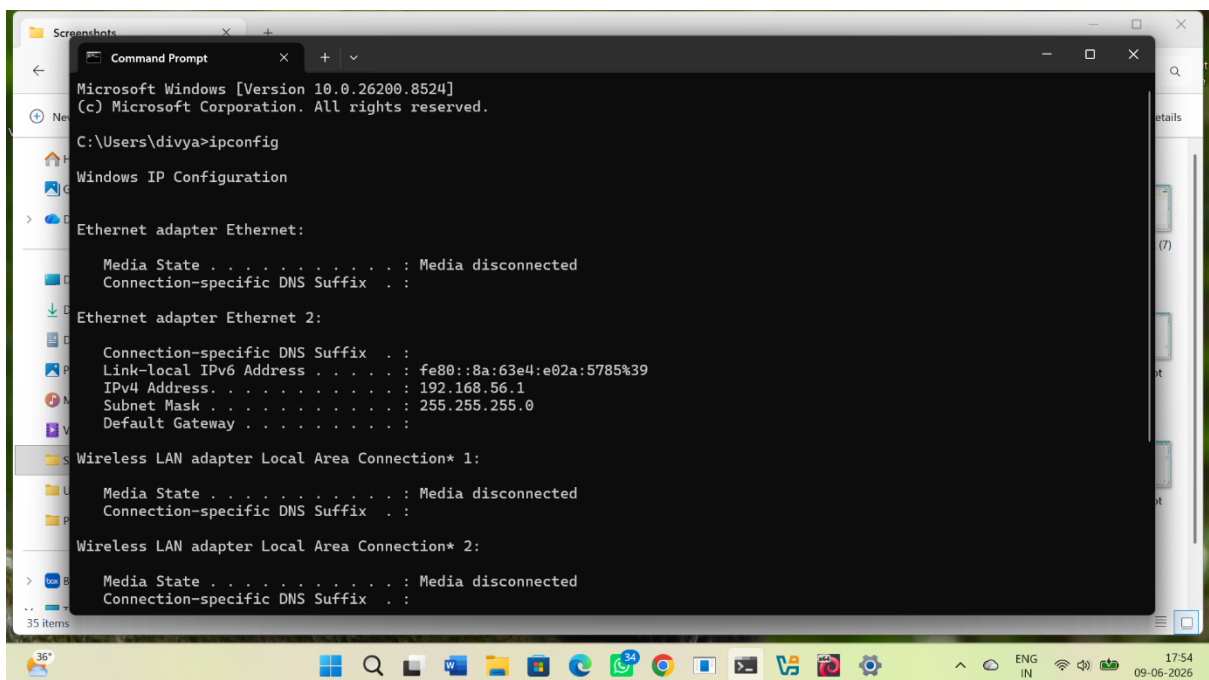
Screenshots captured during the analysis include the following,

Configuring ip addresses

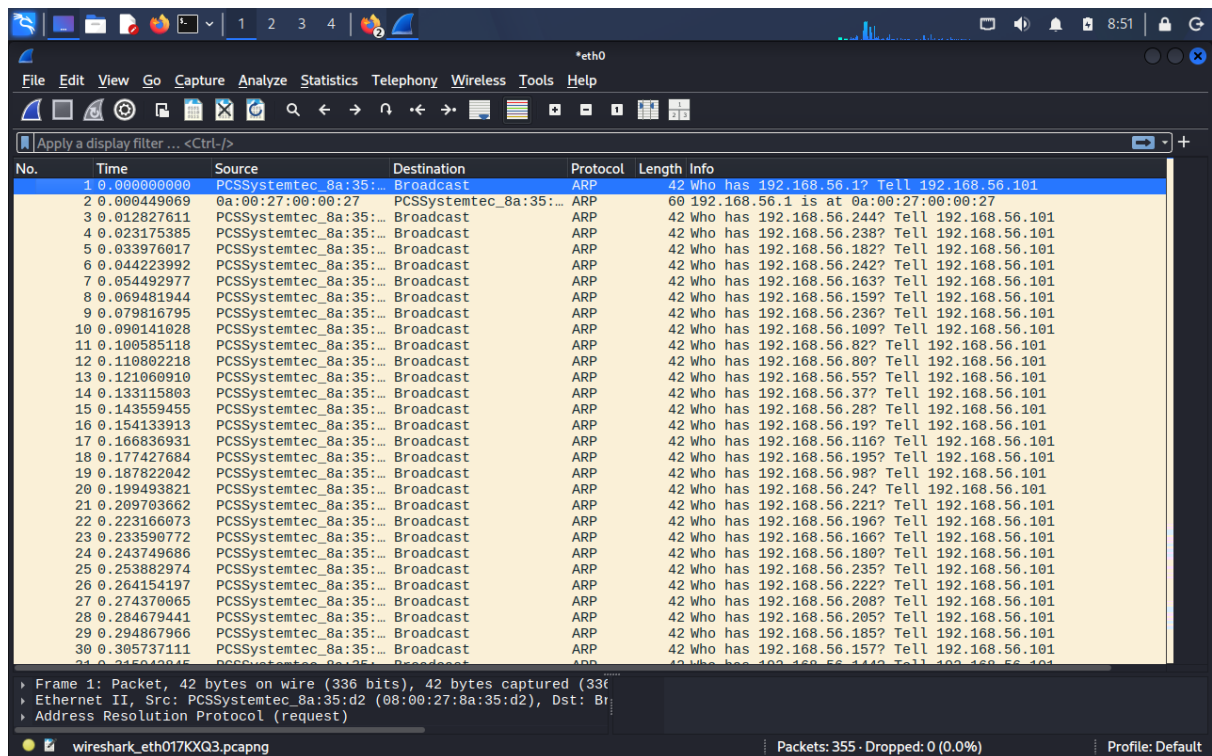
KaliLinux: Ip address: 192.168.56.101



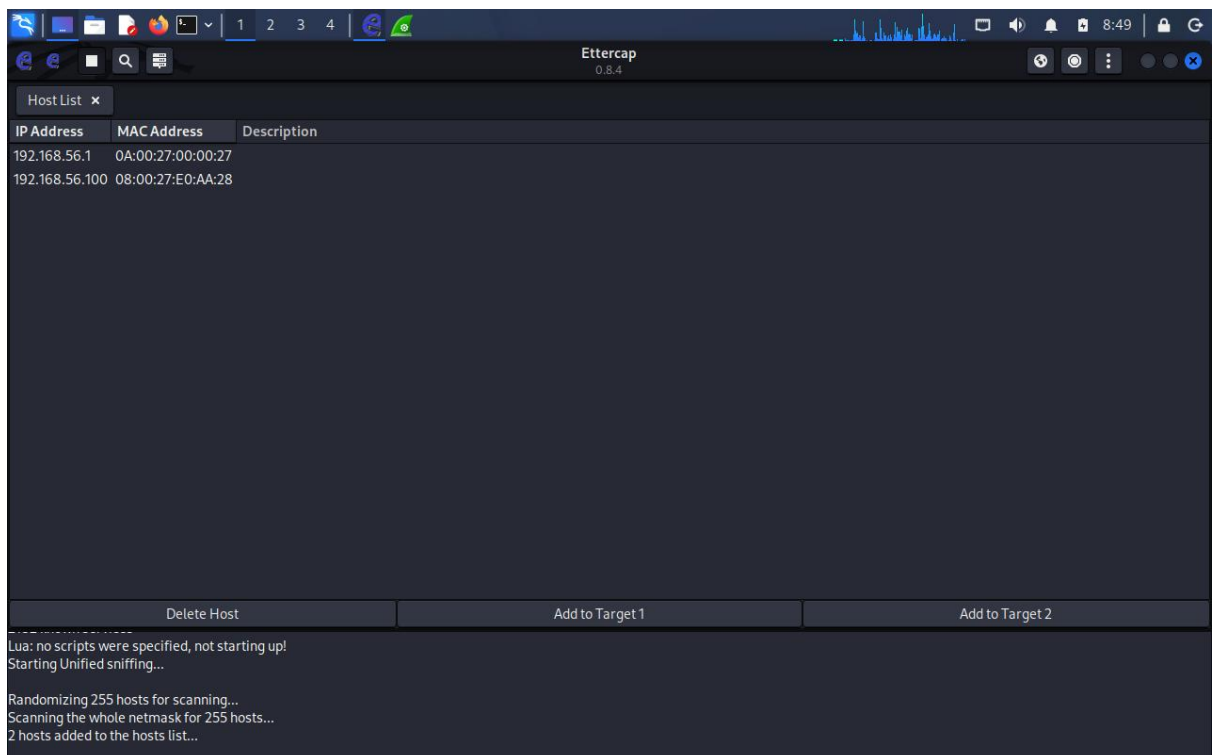
Windows 11: Ip address: 192.168.56.1



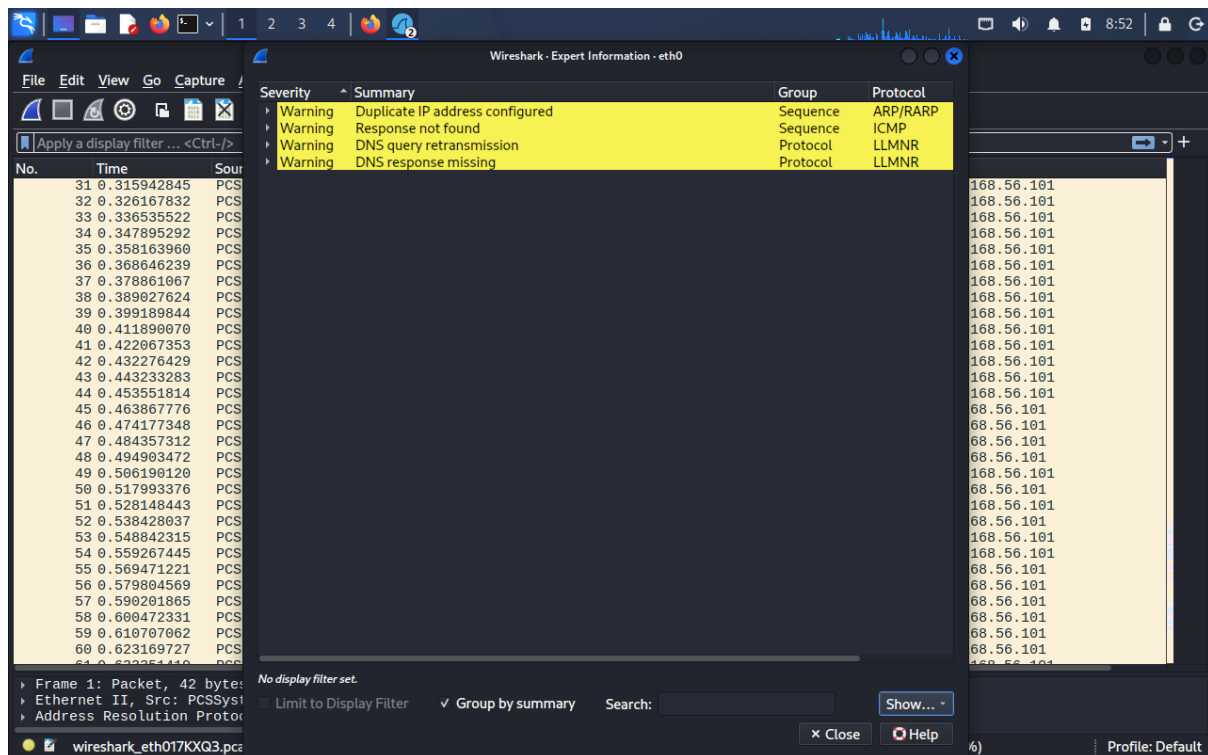
Capturing traffic using Wireshark on Kali Linux



Ettercap Host List



Implimenting wireshark ARP Poisoning and detecting information on Kali Linux



11. Github Link: <https://github.com/divyasri-web/Network-Traffic-Analysis-and-Threat-Detection-Using-Wireshark.git>

Project By,
Divyasri.